

Prototype development of a cross-reality digital twin ecosystem: the web, open source and open data

Wilhelm Herbrich*

Philipp Zittlau

Fabian Joeres

Christian Hansen

Otto-von-Guericke-University Magdeburg,
Research Campus STIMULATE

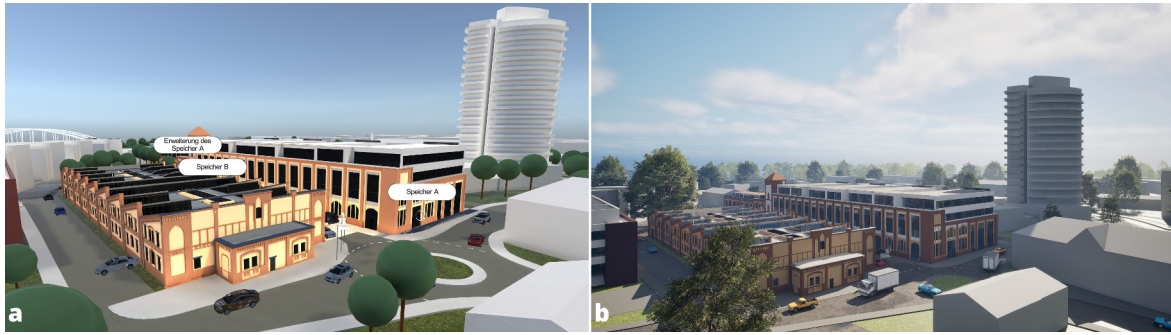


Figure 1: Birds-eye view of a section of prototypes of the digital twin. Visual differences are visible between a) client-side rendering (in-browser) and b) server side rendering (pixel-streaming)

ABSTRACT

This work contributes to a broader initiative aimed at transforming a former industrial port area into a dynamic Knowledge Transfer Space (KTS). To support this transformation, we explore the development of a cross-reality (CR) digital twin of the port area, which integrates user interfaces with varying degrees of virtuality. We evaluate different web technologies, focusing on the balance between accessibility, immersion, scalability, and performance. By comparing client-side rendering with pixel streaming approaches, we aim to identify suitable solutions for prototyping a CR digital twin ecosystem. The development of a prototype is ongoing, based on a client-side rendering approach. The outcomes contribute to developing an open and transferable CR digital twin applicable to similar urban projects in other cities.

Index Terms: Cross reality, Virtual reality, Augmented reality, Knowledge transfer, Smart city, Digital twin, Urban transformation, Science communication

1 INTRODUCTION

Rapid advancements in VR/AR technologies present transformative possibilities for urban areas undergoing redevelopment. This project is part of a multidisciplinary effort to transform a former industrial port into a Knowledge Transfer Space (KTS), blending science, business, and cultural initiatives. In our case, the former river port, though unused for several decades, is situated in a relatively central location of the city. The cities stretch goal is to develop a diverse and lively urban district on the river. A component of participating in this transformation process is the creation of a cross-reality (CR) digital twin to represent the port area digitally, incorporating data such as events, mobility, geographic, and institutional activities.

*e-mail: wilhelm@isg.cs.uni-magdeburg.de

CR, in this context, refers to systems that incorporate user interfaces employing multiple mixed reality modalities, spanning different points along the reality-virtuality continuum [12, 6]. For example, a user may interact with a digital twin with a smartphone, while a different user interacts with the same digital twin with a VR headset. Our goal is to develop a CR ecosystem that engages stakeholders, including local residents, researchers, startups, and businesses, leveraging web technologies and open source solutions.

In this context, open source refers to software whose source code is freely available for anyone to view, use, improve, and distribute [7]. This encourages collaboration and transparency, which makes open source particularly valuable in projects that prioritize reproducibility, accessibility, and community engagement.

This paper evaluates two technological approaches for building a prototype implementation of a digital twin: client-side rendering using the Needle Engine and server-side pixel streaming. These technologies are assessed for their suitability to create an accessible and scalable digital twin, ensuring utility for a broad audience.

The primary research question that guides this study is “What open source technologies can facilitate the creation of an urban digital twin cross-reality ecosystem?” By addressing this question, we aim to provide useful information on building a CR digital twin that can be replicated and adapted for similar projects around the world. As this is a work in progress, the current findings and implementations are intended to inform future developments and iterations of the CR ecosystem.

2 BACKGROUND

2.1 Related work

An important factor in creating urban digital twins is the management of data. The German Federal Institute for Research on Building, Urban Affairs, and Spatial Planning calls data governance a central component of a digital twin [1]. Data quality is crucial, combined with the question of whether data are generally open or closed.

Schrotter et al. built a digital twin of Zurich City using open government data [10]. They identify use cases in city and construction planning, as well as in urban mobility and climate management. By

using open data and standards, they also see good adaption options for further use cases. White et al. introduce a layered architecture for their open digital twin of the Dockland area in Dublin [16]. They use open data sources and offer opportunities for citizens to participate, for example, by providing feedback on urban planning processes.

Another aspect when talking about citizen engagement in urban digital twins is their presentation. VR and AR can offer new ways to participate while providing greater immersion and greater engagement [8, 15]. For example, in a VR environment of a CR system, the user can interact with a completely synthetic world, while in an AR environment of the same CR system, the real environment is enhanced with virtual objects.

When developing a digital CR twin, the technical implementation and concrete use cases must be considered. Stacchio et al. introduced WiXaRd, a modular system that can be used to any degree along the continuum of reality-virtuality – from 2D displays to common VR Head-Mounted Displays (HMDs) [13]. For the second, there are examples of architectural visualization [11] or lighting planning in a public park [9].

2.2 Knowledge transfer space

The presented project is part of a multidisciplinary initiative located in a former industrial river port. The goal of the initiative is to create a KTS, where science findings can be transferred to companies and society. Simultaneously, the area is facing structural change that is intended to transform it into a modern urban neighborhood that combines science, business, and cultural offerings. To increase visibility and accessibility of the initiative, a digital twin is to be developed. Although the term digital twin commonly describes the virtual model of a physical system [14], this term is used more broadly in our case. The digital twin of the KTS is intended to be an ecosystem that represents all the processes that take place there. That can include data like mobility or geographic data, but can expand to activities of companies and research institutions located there, as well making possible changes in the areas infrastructure and appearance more accessible.

2.3 Concept development of a cross-reality ecosystem

To formulate the vision of the KTS digital twin, a requirement analysis was performed according to the user-centered design process [3]. We identified and grouped the needs of stakeholders who came from different backgrounds and whose expertise about VR/AR technologies was limited. To overcome this, a three-step process was implemented:

1. **Stakeholder identification and analysis:** Stakeholders were identified and interviewed using a questionnaire. The results were evaluated using a scoring procedure. This score enables the prioritization of the collected user groups, interaction channels, and content categories.
2. **Selection of technologies:** Eight immersive technologies were cross-referenced with previously identified user groups, interaction channels, and content categories, resulting in a score of their individual compatibility.
3. **Initial use case development:** Technologies were selected based on the scoring and an internal design workshop developed use cases which were defined by the target user groups, the related content, and the selected technologies.

When evaluating the results of Step 2, it became clear that the digital twin should not use just one of the available technologies. Therefore, our concept describes a CR ecosystem that contains three technologies with increasing immersion but decreasing accessibility: a web application, a mobile AR solution using QR codes,

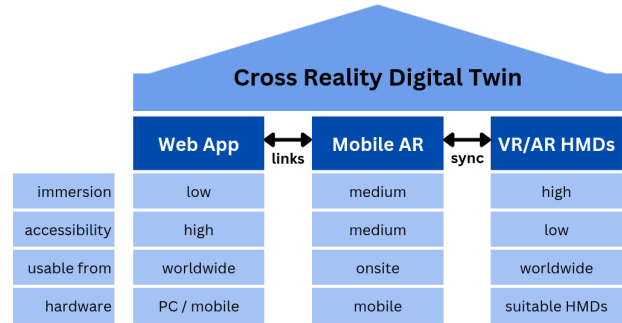


Figure 2: Three pillars visualizing the properties and interaction between the three immersive technologies selected.

and various applications for VR/AR HMDs – see Fig. 2. While a web app and mobile AR were selected due to their overall high score, HMDs were convincing due to their strong rating in the context of events and exhibitions and due to providing high immersion.

Step 3 resulted in three use cases: (1) a *visualization* of status quo and future developments of the real estate situation, (2) the interactive *presentation* of research and development facilities available in the KTS, (3) an *illustration* of historical and cultural contexts. All use cases should be available for all three technologies. To achieve this, data and content must be synchronized between them. However, because of the different interaction techniques, the concepts must be adapted for each technology.

3 METHODS

We present a comparative evaluation of web technologies that we have identified as suitable for implementing our CR use cases. This includes an analysis of client-side rendering using the Needle Engine versus server-side pixel streaming approaches. The comparison focuses on assessing their performance, scalability, immersion, accessibility, and adaptability.

3.1 Web technologies in game engines

After developing use cases and understanding stakeholder needs more precisely, the possibility of facilitating a CR ecosystem became evident. We aim to provide such an ecosystem in an accessible and sustainable manner. Integrating web technologies with game engines is a possible approach to achieve this. This method allows urban digital twin platforms to offer immersive user interactions through web browsers, eliminating the need for specialized hardware or installing apps. We explore two options in this context, Needle Engine and pixel streaming. We evaluated their effectiveness in creating an adaptable urban CR system.

3.2 Needle engine

The Needle Engine¹ is a Unity-based package designed to export content to the web. This allows for the development of common 3D web applications in Unity. The Needle Engine also builds on the WebXR API², enabling the development of immersive CR web applications that are expected to remain accessible. This is due to the API being defined by the industry-backed, open W3C standards body, ensuring backward compatibility, similar to other Web APIs. The major browser vendors, such as Apple Safari, Google Chrome, and Mozilla Firefox, also support WebXR. Furthermore, the Needle Engine attempts to maintain content compatibility using the glTF 3D file format specification³. In-built multi-user in-

¹<https://engine.needle.tools/docs/unity/>

²<https://www.w3.org/TR/webxr/>

³<https://registry.khronos.org/glTF/specs/2.0/glTF-2.0.html>

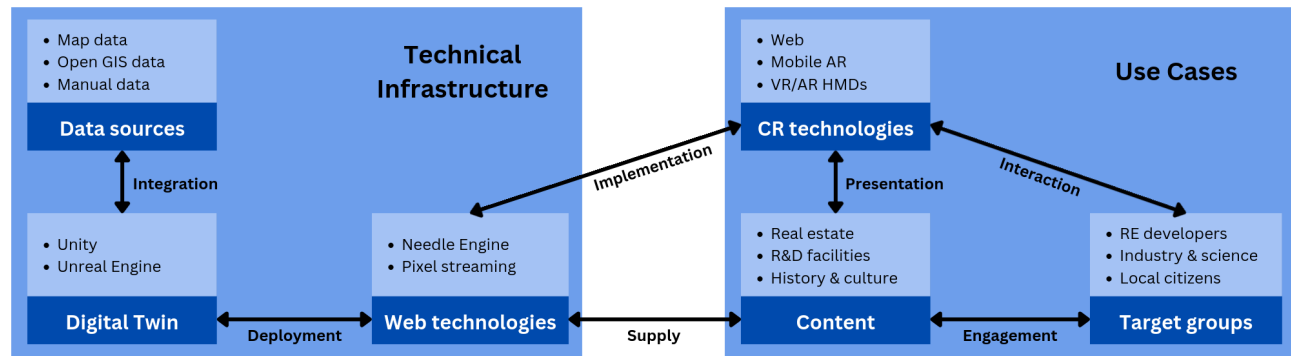


Figure 3: Overview of the integration of technical infrastructure and use case foundations.

interactions and session management capabilities potentially make it suitable for collaborative CR experiences. Additionally, it supports automated level of detail (LoD) generation, improving performance across devices with varying capabilities. The Needle Engine is restricted to the Unity or Blender ecosystems, offering less flexibility for developers who are accustomed to other game engines. Furthermore, complex scenes may encounter low usability due to performance limitations inherent to browsers and rendering the scene on-device.

3.3 Pixel streaming

Pixel streaming enables the delivery of high-quality rendered content from a game engine to any device with a web browser by leveraging server resources [2]. This approach achieves a high LoD and real-time interaction without necessitating high-end hardware on the client. With a sufficiently fast connection to the server, pixel streaming can support low-latency interactions [5], making it well suited for immersive CR applications in digital twin environments. The server renders pixels to be streamed and is in full control of the developer. This leads to a high flexibility in using hardware and software to render the pixels.

As pixel streaming relies on stable, high-bandwidth internet connections, accessibility will be limited in areas with poor connectivity. This is especially critical for AR applications where alignment between real-world and virtual content is dependent on low latency. The continuous transmission of pixel data may also impose unreasonable expenses on users with mobile data limits. Additionally, the use of server resources to render images can lead to higher operational costs, potentially impacting scalability and budget considerations.

Multi-user collaboration via the web necessitates a server to manage user connections. In contrast, a single-user experience rendered client-side, such as with the Needle Engine, can be served by a static web server, significantly reducing maintenance and operational costs. In contrast, pixel streaming services incur higher costs and maintenance requirements, even for single-user experiences.

3.4 Data sources

The accuracy, relevance, and integration of various data sources influences the success of a digital twin ecosystem [1]. Using geospatial data, we aim to enhance realism and provide useful interactivity with the digital twin. By evaluating and potentially combining different data sources, our goal is to provide contextual information tailored to different stakeholders. This section outlines the technologies assessed.

3.4.1 Map data streaming

The integration of live map data tools allows for the creation of realistically accurate representations of urban environments, as these are often based on satellite imagery. Additionally, these services are easy to integrate with WebGL and VR/AR frameworks such as the Needle Engine. We see the possibility of map data streaming as a foundation interacting with the digital twin. We are currently evaluating different services in terms of licensing, data accuracy, and usability for the defined use cases. Additionally, rendering dense datasets may present performance challenges, requiring optimization to maintain usability.

3.4.2 Open GIS data

GIS data (geographical information system) can be ambiguously defined as spatial information about geographic features used for mapping and analysis [4]. Open GIS data leverages open source information that is usable for modeling urban environments. This data can be used across a wide range of applications, such as climate, weather, and hydrological simulation, or to visualize administrative data, such as traffic and infrastructure. We aim to use this data on the foundation of map data to selectively serve specific use cases, e.g. facilitating historical data on buildings to better communicate past purposes to local citizens.

There are potential gaps in the quality and availability of GIS data sets that can limit the accuracy and utility of the models. Furthermore, effective use of this data requires specialized expertise in geospatial data handling and integration, which poses a potential barrier for the implementation of the use cases.

4 RESULTS

We have chosen to focus on the Needle Engine for our architecture due to its fit for creating flexible and immersive web-based CR experiences. Our proposed architecture integrates web technologies to create a CR ecosystem that spans the three modalities described previously: web app, mobile AR, and VR/AR HMDs. The integration of use cases and technologies is illustrated by Fig. 3.

The CR digital twin will provide interactive content, synchronizing data from diverse sources such as live map streams and open GIS data. This multilayered system ensures integration across various devices and offers experiences for diverse stakeholders. Our architecture emphasizes adaptability with the aim of accommodating future expansions and evolving user needs within the KTS. We are currently implementing a first prototype of these solutions to ensure an effective and adaptable system.

5 CONCLUSION AND OUTLOOK

In conclusion, we explore the possibility of using open source and web-based technologies to create a scalable CR digital city twin

platform within the KTS. The adaptability of the Needle Engine makes it a promising tool for developing CR applications that can cater to diverse use cases. Moving forward, we recognize that we are still in the early stages. Our next steps involve implementing the three defined use cases: visualizing current and future real estate developments, showcasing R&D facilities, and depicting historical and cultural narratives. By assessing user engagement and functionality, we aim to refine and enhance the platform. The open source nature of our system allows us to transfer our digital twin technologies to other cities. Urban digital projects could easily build on our solution. This work underscores the potential of open source open data digital twins to facilitate sustainable urban transformation.

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